SS-INR: Spatial-Spectral Implicit Neural Representation Network for Hyperspectral and Multispectral Image Fusion

Xinying Wang, Cheng Cheng, Shenglan Liu, Ruoxi Song, Xianghai Wang, and Lin Feng

***Abstract*— Due to the limitation of imaging equipment, it is difficult to acquire hyperspectral images (HSIs) with high spatial resolution directly. Existing approaches improve the resolution of HSIs by fusing multispectral image (MSI) and HSI. However, most of them are only feed-forward. They only learn low-resolution (LR) to high-resolution (HR) feature mappings without considering the ill-posedness of super-resolution tasks, leading to a large solution space of mapping functions and making it difficult to learn a complete mapping function. Moreover, there is a large resolution difference between HSI and MSI, and some up-sampling operations are inevitably employed in the network. Nevertheless, traditional upsampling methods only represent pixel points in a discrete way, failing to adequately restore the continuous spatial and spectral information. To this end, this article proposes a spatial-spectral implicit neural representation network for hyperspectral and multispectral image fusion (SS-INR). Inspired by the success of implicit neural representation (INR) in continuum reconstruction, we design spatial-INR and spectral-INR for spatial and spectral resolution reconstruction, respectively. SS-INR contains two processes: forward fusion (FF) and back-projection fusion (BPF). In the FF process, the input HSI is first spatially upsampled with spatial-INR to overcome spatial resolution differences while performing initial fusion with MSI. In the BPF process, we explore the spatial and spectral degradation processes and use them as prior knowledge for error correction. Extensive experiments on five public hyperspectral datasets demonstrate the effectiveness of SS-INR, and SS-INR achieves competitive results compared with existing state-of-the-art (SOTA) fusion methods. The source code for SS-INR will be released at https://github.com/wxy11-27/SS-INR.**

***Index Terms*— Hyperspectral image (HSI), image fusion, implicit neural representation (INR), multispectral image.**

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I. Introduction

IN REMOTE sensing imaging, a larger instantaneous field-of-view (IFOV) must be utilized to achieve acceptable signal-to-noise ratio when dealing with narrower spectral bandwidths. This implies that while maintaining the spectral resolution of the imaging, the spatial resolution is inevitably reduced. However, in practical applications of high-precision remote sensing such as target detection [1], change detection [2], [3], and land cover classification [4], [5], both spatial resolution and spectral resolution hold importance. However, we cannot directly obtain HSIs with high spatial resolution due to the incompatibility of spatial and spectral resolution. Under this circumstance, the HSI and MSI fusion technology has arisen as a means of generating HSI with high spatial resolution. It is achieved by integrating the spatial information from MSI with the spectral information inherent to HSI.

The topic of image fusion has been extensively researched prior to the emergence of deep learning. Early techniques typically perform active level measurements in the spatial or transform domain and manually design fusion rules to achieve image fusion [6]. These methods are referred to as traditional fusion methodologies [7], [8]. The classical traditional image fusion methods predominantly involve multiscale transform-based methods [9], sparse representation-based methods [10], subspace representation-based methods [11], saliency-based methods [12], etc. In terms of feature extraction, existing methods do not account for fundamental differences between diverse source images and often utilize the same transform or representation for feature extraction from source images. With regard to fusion design principles, traditional methods are overly reliant on hand-crafted fusion rules and are unable to adapt to intricate fusion scenarios.

In recent years, the enhanced feature representation capabilities afforded by deep learning models have made significant contributions to the advancement of diverse domains in computer vision. Numerous deep learning–based techniques for image fusion have been naturally developed. Several mainstream CNN-based fusion networks are depicted in Fig. 1(a) and (b). It is evident that most of the existing CNN-based fusion methods adopt pixel level and feature level fusion strategies, which are dedicated to solving the problems of feature extraction, feature fusion, and image reconstruction in the fusion process. Masi et al. [13] proposed PNN (CNN-based pansharpening network) is a pioneer in

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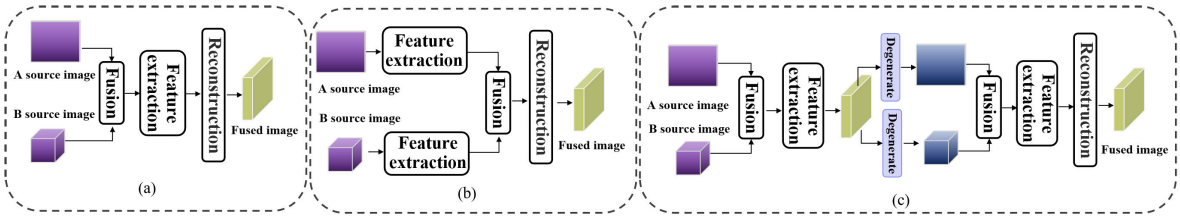


Fig. 1. Different CNN architectures for image fusion. (a) Pixel level fusion. (b) Feature level fusion. (c) DBPN-based fusion.

handling pansharpening (PAN) problems by neural networks, and achieves good reconstruction results with only three layers of convolution under the constraint of l_2 distance loss. Subsequent methods mainly improve the fusion results in terms of both network architecture and constraints. For example, Yang et al. [14] proposed PanNet, which shifts the training to be performed in the high-pass domain by using residual learning, thus allowing the network to focus on the learning of high-frequency structural information, which is more conducive to recovering spatial details. Wang et al. [15] proposed FSL-Unet by combining U-net with an attention mechanism, which achieves spatial-spectral information retention during fusion through dense connectivity and attention mechanism. While the above approach treats the fusion task as a super-resolution problem, TFNet [16] aims to attain PAN and MS image fusion at the feature level. It employs a dual-stream architecture for feature extraction from PAN and MS images, respectively, followed by a network fusion process that integrates the extracted features. The fused features are then inputted into an image reconstruction network to produce the intended high spatial resolution image. HSR-Net [17] combines the channel attention (CA) and spatial attention (SA) modules to enhance the spectral and spatial details of the input image. Additionally, the PixelShuffle module is introduced to preserve the fine details during the upsampling process. Instead of adopting an elaborate network structure, SSR-NET [18] employs spatial and spectral edge loss for cross-modality information fusion, which significantly improves the quality of the fused images.

In summary, the majority of the extant techniques for fusing HSI and MSI are approached from the point of image super-resolution, which gives rise to two problems.

1) Due to the inherent differences in the imaging principles of diverse sensors, the obtained images exhibit diverse resolutions, especially the spectral resolution varies greatly. Thus, integrating the informative content from various source images while overcoming resolution disparities presents a great challenge. Upsampling methods such as deconvolution, bilinear interpolation, among others, are commonly employed in existing fusion networks. These methods represent pixels as discrete points. However, for the human visual system, images are presented in a continuous format. The disadvantage of traditional upsampling methods is that they can only upsample to a fixed scale and cannot achieve super-resolution reconstruction at any scale. Therefore, it limits these methods to extend from the spatial domain to the spectral domain. Thus, we can consider using a continuous function to represent the true state of the image. However, we have no way to know the exact form of this continuous function, so scholars propose to approximate this continuous function by a neural network, which is called implicit neural representation (INR) [19], [20], [21]. Drawing inspiration from the INR approach, we develop an INR-derived upsampling technique, aimed at augmenting spatial and spectral resolution. Different from the previously mentioned upsampling methods, the INR-based upsampling method can better model the continuous representation of spatial and spectral information, which is essential for reconstructing high-resolution (HR) remote sensing images.
2) The current deep learning–based image fusion algorithms mainly address three key issues: feature extraction, feature fusion, and image reconstruction. These methods can be categorized into two distinct classes predicated on their underlying network architectures. The first category involves fusion first, followed by feature extraction, and ultimately image reconstruction, as illustrated in Fig. 1(a). The second category involves separate feature extraction of different source images, followed by feature fusion, and ultimately image reconstruction, as shown in Fig. 1(b). It is noteworthy that these methods adopt unidirectional fusion in designing the network architecture, and do not fully consider the intrinsic relationship between the input data and the fusion results. Instead, they mainly focus on the perspective of image super-resolution. To further investigate the direct interdependence between low-resolution (LR) and HR images, we propose a novel fusion architecture [as shown in Fig. 1(c)] combined with the ideology of deep back-projection network (DBPN) [22]. We explore the spatial and spectral degradation processes from the generative models of HSI and MSI and use them as prior knowledge to guide the optimization of the neural network, which significantly enhances the fusion performance.

To summarize, the main contributions of this article are listed below.

1) We design two components, spatial-INR and spectral-INR, for accomplishing spatial and spectral upsampling, respectively. They set out to more accurately capture the continuous representation of both spatial and spectral information in HSI. To the best of our knowledge, this is the first work that successfully introduces the INR techniques to spectral resolution reconstruction.
2) We propose a novel DBPN-based fusion framework, SS-INR, for HSI and MSI fusion. Different from traditional feed-forward fusion (FF) networks, SS-INR

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leverages the spatial and spectral degradation processes as prior knowledge to guide the optimization of neural networks. This fusion-before-degradation network structure enables a deeper exploration of the interdependent relationship between HR and LR images, which is beneficial for improving fusion performance.

The rest of the paper is organized as follows. Section II presents a review of related work, with a primary focus on INR and DBPNs. Section III provides a detailed description of the proposed SS-INR model. Section IV conducts a thorough analysis of experimental results obtained by comparing the performance of the SS-INR model with that of nine state-of-the-art (SOTA) algorithms on five hyperspectral datasets, including ablation experimental results. Finally, Section V presents the concluding remarks.

II. Related Work

In this section, we will initially present a comprehensive overview of the INR. Subsequently, we will introduce the DBPNs.

A. Implicit Neural Representation

INR is a new method for parameterizing various signals. Traditional signal representations are usually discrete. In the case of images, for example, the most prevalent representation is discrete pixel points in a 2-D space. However, in the real world, we can consider seeing the world as continuous, or approximately continuous. In the case of HSI, it is continuous, narrow-band image data with high spectral resolution. In this way, a continuous function can be deemed to more accurately embody the authentic state of an image. As it is impossible to ascertain the exact form of this continuous function, the concept of an INR has been introduced.

An INR parameterizes the signal as a continuous function, with the objective function commonly realized as a multilayer perceptron (MLP). This MLP serves to map the domain of the signal to the values of the attributes at that coordinate. INR has been widely used in representing various complex signals, such as 3-D computer vision [23], [24], [25], [26], [27], 2-D image synthesis [20], [21], [28], and other fields. INR is a continuous representation process, thus the memory required to parameterize the signal is independent of the spatial resolution and only related to the complexity of the underlying signal. Based on this, Chen et al. [20] proposed a method for learning continuous image representations using local implicit functions, called LIIF. The continuous representation learned by LIIF is capable of interpolating images at arbitrary resolutions, even up to 30× interpolation. Feng et al. [29] propose a point cloud continuous implicit representation that enables prediction of potential surfaces of point clouds, and upsampling at arbitrary resolutions, achieving stunning results in point cloud upsampling tasks. In the field of image generation, previous attempts at INR have been constrained to MNIST-like datasets and could not be extended to complex real-world data. Skorokhodov et al. [21] proposed the INR-GAN architecture, which greatly improved the performance of continuous image generators and greatly reduced the gap between continuous and pixel-based image generators. The CoordConv proposed by Liu et al. [30] uses pixel coordinates as additional input to the neural network to introduce spatial location-related bias. Anokhin et al. [28] proposed an image generation algorithm based on INR to address the problem that standard convolution is still used as the main synthesis operation in the model and the synthesis process of neighboring pixels is not independent, which has achieved good results.

It is well known that HSI has many spectral bands, which are obtained by continuous imaging in a certain spectral range. So the spectral information of HSIs also has a certain continuity. We are greatly inspired by the powerful spatial information characterization capability shown by INR in the fields of image super-resolution and image reconstruction. It is can be more beneficial to use INR for spatial resolution enhancement of remote sensing images and to extend the INR method from spatial dimension to spectral dimension to recover HSIs with arbitrary spectral resolution.

B. Deep Back-Projection Networks

Deep back-projection is an effective iterative method to minimize reconstruction loss. Irani and Peleg [31] pioneered the use of iterative back-projection to solve the image scaling problem. It is one of the most classical algorithms for image super-resolution. The HR image is obtained by iterative back-projection of the errors of the simulated LR image and the observed LR image. Some previous studies [32], [33], [34], [35] have also confirmed the effectiveness of the back-projection method in the field of super-resolution. In recent years, deep learning–based methods have shown excellent performance in super-resolution tasks with their powerful feature representation. The deep-learning-based image super-resolution approaches have learned the representation of image information from LR images first, and then mapped such a representation to HR images by nonlinear mapping, which does not fully use the relationship shared between LR and HR images well. Haris et al. [22] proposed a DBPN for image super-resolution by combining back-projection with deep learning methods. DBPN constructs a network that constantly upsamples and downsamples each other to perform error feedback and achieves good results in large-scale super-resolution reconstruction tasks. Xu et al. [36] proposed a depth-gradient projection network for PAN, which transformed the generation of pan images and MSIs into two optimization problems, and obtained excellent fusion results by iteratively solving these two optimization problems using the gradient projection method.

Inspired by the success of the DBPN in super-resolution and PAN tasks, we venture to introduce it for HSI and MSI fusion. Different from the previous structures, we use a two-branch structure to learn the mapping relationships from HR to LR in spatial and spectral dimensions, respectively. Furthermore, we enhance the performance of our model by incorporating residual computation into the network and utilizing it as a prior knowledge to guide the network optimization.

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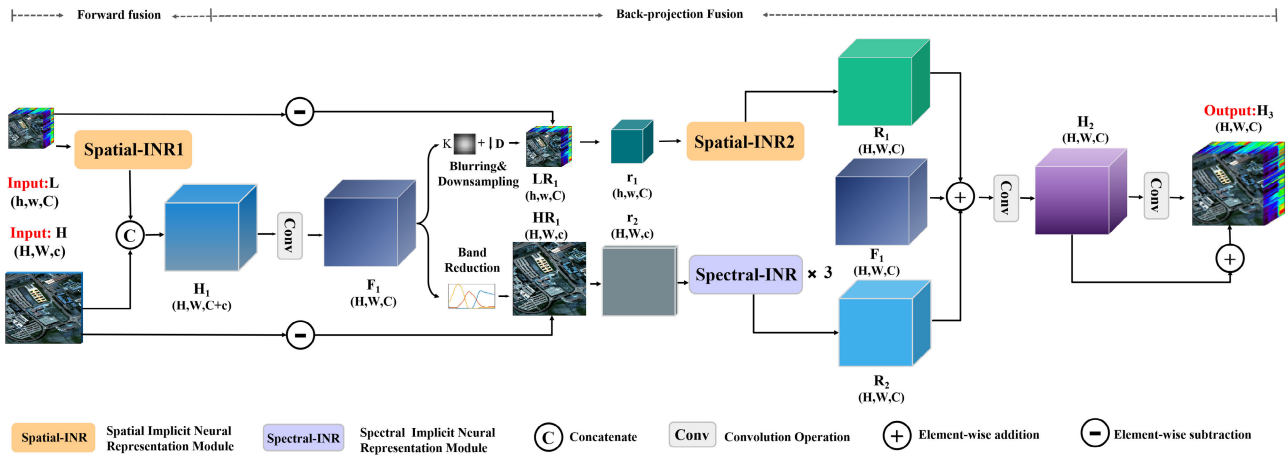


Fig. 2. Overall structure diagram of the proposed SS-INR. The main building blocks of the proposed SS-INR architecture are two parts: FF and BPF.

III. Proposed Method

Fig. 2 shows the overview architecture of the proposed SS-INR. The main building blocks of the proposed SS-INR architecture consist of two parts: FF and back-projection fusion (BPF). The FF process is responsible for the initial fusion of HSI and MSI. we denote the input MSI and HSI as $H \in \mathbb{R}^{H\times W\times c}$ and $L \in \mathbb{R}^{h\times w\times C}$, respectively. First, the input L undergoes upsampling via spatial-INR and is subsequently fused with the H to generate a preliminary fused image F_1. To better learn the features of HR and LR image pairs and to mitigate the noise introduced during the upsampling process, the F_1 is fed into the BPF. Then, the BPF leverages spatial and spectral degradation processes to inform the optimization of the neural network by serving as prior knowledge. Residual calculations are introduced in BPF and the residual features are upsampled by spatial-INR and spectral-INR. It is worth noting that, due to the excessive difference between HSI and MSI spectral resolutions, we use three spectral-INR components in the network for progressively increasing the number of bands. Finally, HR residual feature maps are fused with the initial fusion image to achieve error correction.

A. Forward Fusion

The FF network is designed to primarily fuse HSI and MSI. Accounting for the large spatial resolution differences between them, upsampling operations are required in the network. Traditional upsampling methods, such as interpolation and deconvolution, fail to consider the continuity of spatial information, which may lead to image distortion at HR. Drawing inspiration from the success of image reconstruction networks based on INR, we design a spatial-INR module for spatial upsampling of HSI. The upsampled HSI is subsequently fused with MSI to generate the preliminary fused image.

1) Spatial-INR: Fig. 3 illustrates the overall architecture of spatial-INR. In spatial-INR, the input continuous HSI is represented as a 2-D feature map $\text{lr} \in \mathbb{R}^{H/4\times W/4\times C}$. Assuming that the $H/4 \times W/4 \times C$ feature vectors (also called latent codes) in lr are uniformly distributed in the image domain X. A decoding function f_θ can be used to map the feature lr and the coordinates coord of the continuous domain to the predicted values. The decoding function f_θ can be parameterized as an MLP, which can be expressed as

$$\text{hr}_{\text{spatial}} = f_\theta(\text{lr}, \text{coord}) \tag{1}$$

where lr is the input feature vector, coord is a 2-D coordinate in the continuous image domain, and $\text{hr}_{\text{spatial}}$ is the pixel prediction value at the corresponding coordinate.

a) Cell decoding: For an image reconstruction task with a given resolution, the most straightforward way is to find the corresponding pixel value based on the pixel center coordinates. However, this approach does not consider the real size of the pixel and results in the loss of all other information, except for the pixel center location. To address this issue, spatial-INR uses the cell decoding strategy, which involves incorporating the height and width of the corresponding image block pixels into the network as supplementary information. When the physical size of an image is constant, the smaller the size of the pixels that make it up, the more continuous it is in reality. The f_θ in (1) can be reformulated as f_{cell}, and the above process can be defined as:

$$\text{hr}_{\text{spatial}} = f_{\text{cell}}(\text{lr}, [\text{coord}, d]) \tag{2}$$

where $d = [d_h, d_w]$ denotes the height and width of the current query pixel and [.] refers to the concatenation.

b) Feature expansion: In order to enrich the information contained in the latent codes of lr, we generate lr^* by feature expansion. Each latent code in lr^* is a concatenation of 3×3 adjacent latent codes in lr. The feature vector at position (i, j) after feature expansion can be defined as

$$\text{lr}^*_{ij} = \text{Concat}\left(\left\{\text{lr}_{i+m,j+n}\right\}_{m,n\in\{-1,0,1\}}\right) \tag{3}$$

where lr^*_{ij} denotes the feature vector at position (i, j) after feature expansion, Concat refers to concatenation.

c) Pixel estimation: Fig. 4 depicts the process of pixel estimation. Assuming that X is a continuous image domain and the $H/4 \times W/4$ latent codes in lr^* are uniformly distributed in X, the pixel values at coordinate y can be defined as

$$X(y) = f_\theta\left(a, y - a^{\text{coord}}\right) \tag{4}$$

where a is the latent code which nearest to y in the image domain X and a^{coord} is the coordinate of the latent code a in the image domain X.

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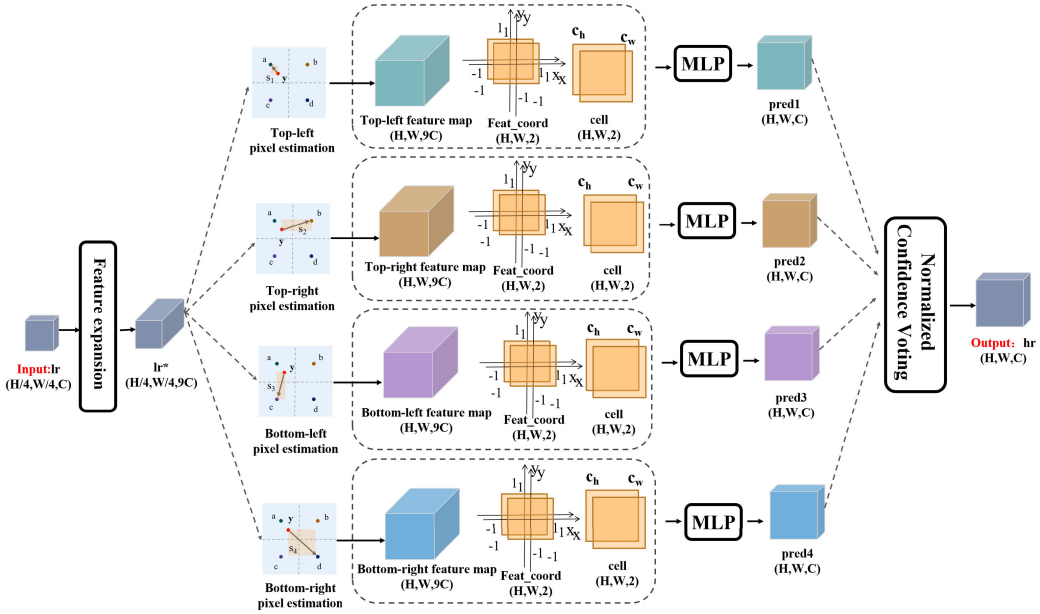


Fig. 3. Overall architecture of spatial-INR. In spatial-INR, the input feature maps are pixel-estimated in four directions after feature expansion. The decoding function maps the features and coordinates in the continuous domain to the predicted values. Finally, the fusion result is obtained by normalized confidence voting.

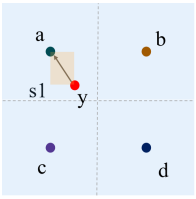


Fig. 4. Schematic of pixel estimation in spatial-INR. The y denotes the coordinates to be estimated, while a represents the closest latent code.

When we make a prediction of signal y based solely on the nearest latent codes in domain X, any movement in the coordinate position of y could result in an abrupt change of its nearest neighbor latent codes. Such a change could lead to discontinuity in the prediction outcome. To address the potential issue of boundary mutation during the pixel estimation process, the spatial-INR employs a multidirectional prediction strategy (as illustrated in Fig. 3), where the value of y is predicted using feature vectors in four directions.

d) Normalized confidence voting: Based on the aforementioned approach, we can obtain four feature maps, each corresponding to a specific direction. As the latent codes in distinct directions are located at varying distances from the predicted pixels, they exert dissimilar influences on the resultant images. For this reason, we use the normalized confidence voting mechanism, which uses the ratio of the area as the weight to sum up the predicted images of the four directions to get the final result. The process can be elucidated as follows:

$$X(y) = \sum_{m\in\{00,01,10,11\}} \frac{s_m}{s} f_\theta\left(a_m, y - a_m^{\text{coord}}\right) \tag{5}$$

where a_m denotes the nearest latent code in the top-left, top-right, bottom-left, and bottom-right sub-spaces, a_m^{coord} is the coordinate of a_m, and s_m is the area of the rectangle between y and a_m^{coord}. s is the total area of the entire region used for normalization.

2) Preliminary Fusion: For the input HSI $L \in \mathbb{R}^{h\times w\times C}$, we upsample it by spatial-INR to eliminate the spatial resolution difference with the input MSI $H \in \mathbb{R}^{H\times W\times c}$. Next, the two images are fused by convolution to obtain the preliminary fused image. The initial fusion process can be described as

$$F_1 = \text{Conv}\left(\text{Concat}\left(\text{INR}_{\text{spatial}}(L), H\right)\right) \tag{6}$$

where Conv$(\cdot)$ is a 3×3 convolution operation, Concat is concatenate and $\text{INR}_{\text{spatial}}$ represents spatial-INR.

B. Back-Projection Fusion

Many fusion methods do not take into account the intrinsic relationship between input data and fusion results when designing the network architecture. Furthermore, the fusion process is typically conducted from a single perspective of image super-resolution. To address this issue, we propose the BPF component, which combines the generative models of HSI and MSI with a DBPN. Specifically, BPF includes degradation and reconstruction processes. The spatial and spectral degradation processes are first explored from the generative models of HSI and MSI, and the residuals are calculated from the degraded images and the original input images, which are used as prior knowledge to guide the optimization of the network. The reconstruction process adopts a two-branch structure, containing spectral and spatial branches. The residual feature maps are upsampled in spatial dimension and spectral dimension respectively. The spatial upsampling is completed using the spatial-INR introduced in Section III-A1. For spectral upsampling, we design the spectral-INR component to simulate the continuous representation.

1) Degradation Process: Since reconstructing an HR-HSI is an ill-posed problem, a number of possible HR images may

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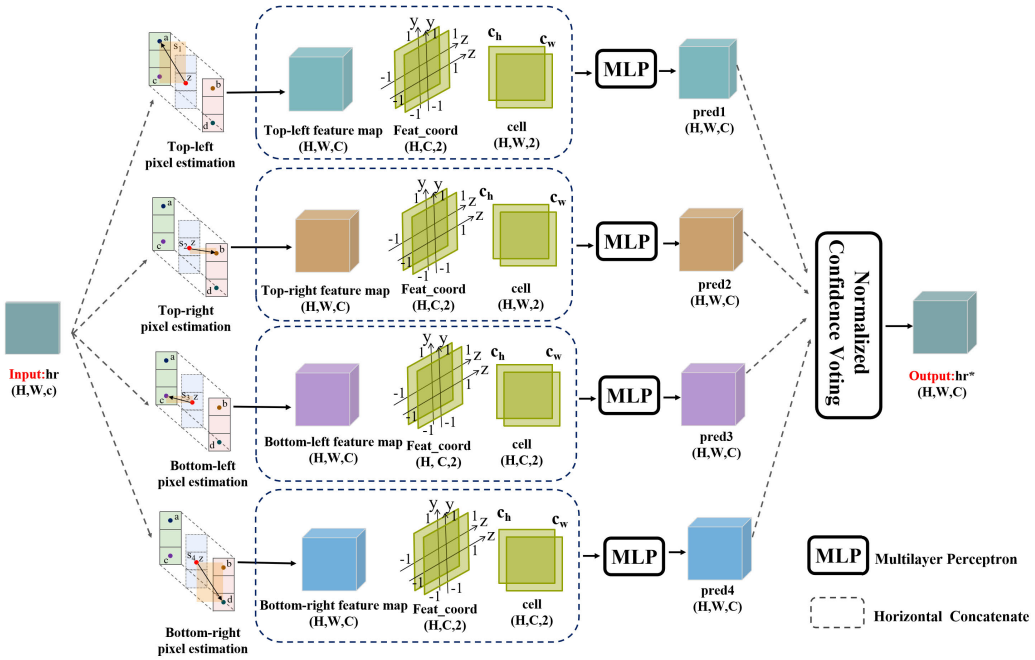


Fig. 5. Overall architecture of spectral-INR. In spectral-INR, the decoding function MLP maps the features and coordinates in the continuous spectral domain to the predicted values. Finally, the fusion result is obtained by normalized confidence voting.

exist that could be downsampled to produce the same LR image [37], which will result in a large mapping space from LR to HR. Based on this, we simulate the spatial and spectral degradation process and use it as prior knowledge to guide the optimization of the neural network, so as to decrease the size of the solution space and improve the network performance. Specifically, for the initially fused image F_1, we first generate LR_1 and HR_1 according to the degradation model, and this process can be described as

$$\begin{aligned} \mathrm{LR}_1 &= DKF_1 \\ \mathrm{HR}_1 &= F_1 S \end{aligned} \tag{7}$$

where $D \in \mathbb{R}^{HW \times hw}$ denotes a downsampling matrix and K is the circular convolution matrix, and $S \in \mathbb{R}^{C \times c}$ is the spectral response function.

We argue that if F_1 is an accurate enough fusion result, then the LR_1 and HR_1 generated by F_1 via the degradation model should be similar enough to the original input of L and H. Therefore, we introduce residual computation in the network and subsequently re-added the residual feature map as a prior information to implement feature compensation in the feature map. Define the residual feature maps r_1 and r_2 as

$$\begin{aligned} r_1 &= L - \mathrm{LR}_1 \\ r_2 &= H - \mathrm{HR}_1. \end{aligned} \tag{8}$$

Spatial and spectral resolution enhancement was applied to the two residual feature maps r_1 and r_2 using spatial-INR and spectral-INR, respectively. The spatial-INR process is detailed in Section III-A1, and the spectral-INR process is elaborated below.

2) Spectral INR: The network structure diagram of spectral-INR is shown in Fig. 5. Since the spectral curves obtained by HR sensors are continuous spectral signals, this means that HSI and MSI are also continuous in the spectral domain similar to the spatial domain. We define $\mathrm{hr} \in \mathbb{R}^{H \times W \times c}$ as an input feature map with high spatial resolution and low spectral resolution, uniformly distributed in the 2-D continuous image domain $X \in \mathbb{R}^{H \times W \times c}$. We define a decoding function g_θ to predict the pixel values of missing bands, whose parameters are shared by the entire image. g_θ is an MLP that can be described by

$$\mathrm{hr}_{\mathrm{spectral}} = g_\theta(\mathrm{hr}, \mathrm{coord}) \tag{9}$$

where coord is a 2-D coordinate in the continuous image domain, and $\mathrm{hr}_{\mathrm{spectral}}$ is the pixel prediction value at the corresponding coordinate.

a) Cell decoding: Similar to spatial resolution enhancement methods, the most common method used in spectral resolution enhancement tasks is to predict the corresponding values based on the center coordinates of the pixel points. However, this approach does not take into account the spectral bandwidth, which means that all other information about the pixel point, except the center position, is lost. To address this drawback, we propose a cell decoding strategy, where the height and bandwidth of pixel points in the spectral domain are provided to the network as supplementary information, which can be expressed as

$$\mathrm{hr}_{\mathrm{spectral}} = g_{\mathrm{cell}}(\mathrm{hr}, [\mathrm{coord}, e]) \tag{10}$$

where $e = [h, w]$ denotes the height and width of the currently estimated pixel in the spectral domain.

b) Pixel estimation: Fig. 6 depicts the process of pixel estimation in spectral-INR. Assuming that Y is a continuous spectral domain and the latent codes in hr are uniformly distributed in Y, the pixel reconstruction process at coordinate x is

$$Y(x) = g_\theta\left(a, x - a^{\mathrm{cood}}\right) \tag{11}$$

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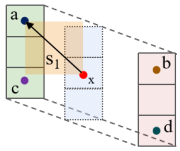


Fig. 6. Schematic of the pixel estimation process in spectral-INR. The x denotes the pixel to be estimated, while a is one of its nearest latent codes.

Algorithm 1 Spectral INR

Input: Feature map hr with low spectral resolution.

1: The pixel prediction is performed in four directions via Eq. 11 to obtain the top-left feature map, top-right feature map, bottom-right feature map, and bottom-left feature map, respectively.
2: Horizontal concatenate the feature map, coordinates, and cell information in the corresponding direction.
3: Generate four prediction feature maps via Eq. 10.
4: Generate final result by the normalized confidence voting strategy via Eq. 12.

Output: Feature map hr^* with high spectral resolution.

where a denotes the latent code in spectral domain Y that is closest to x, where both a and c are equidistant from x, and a is selected as the representative. Further, a^{cood} is the coordinate of the latent code a.

The prediction of the signal x in (11) is obtained by querying the latent code b in the spectral domain that is nearest to it. In the spectral dimension X, when x moves, the boundaries of a change abruptly, resulting in discontinuous prediction results. To solve this problem, we adopt the same strategy as in spatial-INR, that is, predictions are generated in four directions in the spectral dimension (as illustrated in Fig. 5). Subsequently, the forecast outcomes from each direction are subjected to weighting and averaging based on distance, to facilitate a smooth transition across the boundary region.

c) Normalized confidence voting: Since the distance between the latent code and the pixel to be estimated is different in the four directions, the impact on the result is also different. So for these four feature vectors, the ratio of the area is used as the weight, and the fusion is performed using the normalized confidence voting mechanism, and the final result can be expressed as

$$Y(x) = \sum_{m\in\{00,01,10,11\}} \frac{s_m}{s} g_\theta\left(b_m, x - b_m^{\text{coord}}\right) \tag{12}$$

where b_m is the nearest latent codes in the top-left, top-right, bottom-left, and bottom-right sub-spaces, b_m^{coord} is the coordinate of b_m, and s_m is the area of the rectangle between x and b_m^{coord}. s is the total area of the entire region used for normalization.

Adhering to all the implementations proposed above, the proposed spectral-INR can be summarized as in Algorithm 1.

3) Final Fusion: By spatial-INR and spectral-INR to upsample the residual feature maps r_1 and r_2, respectively, we obtain the HR feature maps R_1 and R_2. R_1, R_2, and the initially fused image F_1 are summed for deep feature fusion. After the above process, we obtain the fused feature map H_2, and then further reconstruct the spatial and spectral information by a convolution layer. The final fusion result H_3 can be defined as

$$\begin{aligned} H_2 &= \text{Conv}(R_1 + R_2 + F_1) \\ H_3 &= H_2 + \text{Conv}(H_2) \end{aligned} \tag{13}$$

where $\text{Conv}(\cdot)$ represents a 3×3 convolution operation.

C. Loss Function

In this study, the mean squared error (mse), a simple and widely adopted loss function, was adopted. The network achieved favorable convergence without imposing excessive constraints. The loss function L can be defined as

$$L = \frac{\sum_{i=1}^{H}\sum_{j=1}^{W}\sum_{k=1}^{C}\left(\left(\hat{Z}(i,j,k) - Z(i,j,k)\right)^2\right)}{2WHC} \tag{14}$$

where $Z \in \mathbb{R}^{H\times W\times C}$ is the reference HR-HSI and $\hat{Z} \in \mathbb{R}^{H\times W\times C}$ denotes the reconstructed HR-HSI.

IV. Experimental Results and Analysis

This section is mainly dedicated to presenting the dataset description (Table I provides comprehensive details on the datasets) and evaluation metrics employed in our experiments. It also includes a discussion of the experimental setup. Subsequently, we conducted experiments on five hyperspectral datasets (four simulated datasets and one real dataset) and compared the results with nine fusion methods. In addition, ablation experiments were performed on each individual module within the network, and the temporal efficiency of the network was further explored.

A. Datasets Description

1) Pavia Centre: The Pavia Centre dataset was obtained using the ROSIS sensor, comprising an original dataset of 115 bands. In this study, 102 of these bands were selected following the removal of factors such as water vapor and noise. The spatial resolution of the Pavia Centre dataset is 1.3 m, and it encompasses nine feature classes.

2) Botswana: The Hyperion sensor on the EO-1 platform captured the Botswana dataset with a 30-m pixel resolution. The sensor acquired 242 bands, of which 145 bands, after eliminating the uncalibrated and noisy bands, were utilized for this study.

3) Houston: The Houston dataset was obtained through the ITRES CASI-1500 sensor, courtesy of the 2013 IEEE GRSS Data Fusion Competition. The dataset possesses a spatial resolution of 349×1905 and encompasses a spectral range from 364 to 1046 nm, with a total of 144 bands.

4) Pavia University: Similar to the Pavia Centre dataset, the Pavia University dataset was also acquired by the ROSIS sensor. The sensor contains a total of 115 bands, and after processing, we use 103 of these bands. The size of Pavia University is 610×340.

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TABLE I
DATASET DESCRIPTION

Dataset	Pavia Center	Botswana	Houston	Pavia University	Xiongan
Sensor	ROSIS	EO-1	CASI-1500	ROSIS	GF Special Program
Location	Pavia Center, Italy.	Okavango Delta, Botswana.	University of Houston, USA	Pavia University, Italy	Horseshoe Bay Village
Image Size	1096×1096	1476×256	349×1905	610×340	1500×3000
Spatial Resolution	1.3 m	30 m	2.5 m	1.3 m	0.5 m
Bands	102	145	144	103	256

5) Xiongan: The dataset of the Xiongan includes HR-HSI, LR-HSI, and RGB images, which were collected by the GF specialized aerial system multispectral imaging spectrometer. The HR-HSI spectral range is 400–1000 nm, with 256 bands, and the image size is 1500 × 3000 pixels, with a spatial resolution of 0.5 m. LR-HSI has an image size of 500 × 1000 and a spatial resolution of 1.5 m.

For the experiments on the simulated dataset, we are generating two input images according to Wald's protocol [38]. The original HSI served as the reference image. Define L as the input HSI, which is obtained from the reference image after Gaussian blurring and downsampling by a factor of four. The input MSI denoted as H is composed of five bands extracted from the reference image at equal intervals. A sub-region of the data center with dimensions of 128 × 128 pixels is selected for testing, while the remaining data is allocated for training. To ensure non-overlapping regions between the training and testing samples, the test region of the image samples is filled with zeros during model training. The training region is randomly cropped to a size of 128 × 128 pixels for each iteration during training. For the real dataset, a sub-region of the data center with a size of 126 pixels was selected for testing, while the remaining data is allocated for training.

B. Evaluation Metrics

In order to conduct a quantitative assessment of the performance of SS-INR, four evaluation metrics are introduced in this study, namely root-mean-squared error (RMSE), peak signal-to-noise ratio (PSNR), Erreur Relative Globale Adimensionnelle de Synthèse (ERGAS), and spectral angle mapper (SAM). To this end, we define $F_k(i, j)$ to denote the reference HSI at the k-band and position (i, j), while $\hat{F}_k(i, j)$ is used to denote the estimated HSI at the same K-band.

1) Root-Mean-Squared Error: RMSE is utilized to quantify the discrepancy between two images, where a lower RMSE value indicates a stronger resemblance of the fusion outcome to the original image at the pixel level, hence implying the superior performance of the model. Mathematically, the RMSE can be expressed as follows:

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{k=1}^{C}\sum_{i=1}^{H}\sum_{j=1}^{W}\left(F_k(i, j) - \hat{F}_k(i, j)\right)^2}{HWC}} \tag{15}$$

here, c represents the number of channels of the image, h represents the height, and w represents the width.

2) Peak Signal-to-Noise Ratio: PSNR measures the proportion of peak power to the noise power in the fused image, providing insights into the level of distortion occurring during the fusion process. Higher values of PSNR indicate lesser amounts of distortion, and the fusion results have better spatial detail information. The PSNR of ith band is defined as

$$\mathrm{PSNR} = 10\log_{10}\left(\frac{\max(F_i)^2}{\frac{1}{HW}\left\|F_i - \hat{F}_i\right\|_2^2}\right) \tag{16}$$

here, H and W represent the height and width of the image, respectively. $\|\cdot\|_2$ is the two-norm.

3) Erreur Relative Globale Adimensionnelle de Synthèse: ERGAS is a comprehensive metric to assess the average error and dynamic range variation between the fused result and the reference HR-HSI. A reduced value of the ERGAS metric signifies a higher degree of similarity between the fused image and the reference image, and it can be defined as

$$\mathrm{ERGAS} = 100\frac{h}{l}\sqrt{\frac{1}{B}\sum_{k=1}^{B}\left(\frac{\mathrm{RMSE}(H_k, F_k)}{\mu_k}\right)^2}. \tag{17}$$

Here, where (h/l) is the ratio of the resolution of MSI and HSI, μ_k denotes the mean value of the kth band of HSI, while H_k denotes MSI, with B denotes the number of bands.

4) Spectral Angle Mapper: SAM assesses the spectral accuracy of the fusion result by measuring the angle between the reference and fused images. A lower value of SAM indicates successful reconstruction of spectral information during the fusion process. The formula for computing SAM is presented below

$$\mathrm{SAM} = \arccos\left(\frac{\left\langle F(i, j), \hat{F}(i, j)\right\rangle}{\|F(i, j)\|_2\left\|\hat{F}(i, j)\right\|_2}\right). \tag{18}$$

Here, $\langle\cdot,\cdot\rangle$ means inner product operation. The final SAM results are obtained by averaging the sum of SAM values for each pixel.

C. Implementation Details

SS-INR is implemented by Python 3.7 under the Pytorch 1.2.0 framework. We trained the model on one GeForce RTX 3090 GPU and used the Adam optimizer. We adopt a learning rate decay strategy. Specifically, the initial learning rate was set to 1e-3, and it was reduced by a factor of 0.5 every 2000 epochs.

D. Comparison With SOTA Methods

We conducted a comparative analysis of the SS-INR method with ten SOTA techniques on five popular hyperspectral

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TABLE II
EXPERIMENTAL RESULTS ON THE PAVIA CENTRE DATASET, WITH THE FIRST AND SECOND PLACE BEING, RESPECTIVELY, MARKED IN **RED** AND **BLUE**

Method	Pavia Center			
	RMSE	PSNR	ERGAS	SAM
ICCV15	5.0793	34.4197	5.6044	5.8472
FUSE	4.1070	37.7270	4.8480	4.5984
MSDCNN	4.1687	35.7370	4.6840	5.4004
TFNet	4.2473	35.5686	4.8295	4.8437
ResTFNet	3.9258	36.2522	4.4449	4.6073
SSFCNN	4.7720	34.5569	5.7509	5.7962
ConSSFCNN	5.4998	33.3239	6.1594	6.1196
SSR-NET	3.4102	37.4752	3.8964	3.8618
HSR-NET	3.4048	37.4889	3.8402	4.1796
MCT-Net	**2.7520**	**39.3378**	**3.0751**	**3.6897**
SS-INR	**2.5745**	**39.9169**	**2.9595**	**3.4244**

TABLE III
EXPERIMENTAL RESULTS ON THE BOTSWANA DATASET, WITH THE FIRST AND SECOND PLACE BEING, RESPECTIVELY, MARKED IN **RED** AND **BLUE**

Method	Botswana			
	RMSE	PSNR	ERGAS	SAM
ICCV15	1.4904	20.6337	5.5331	5.8677
FUSE	1.0811	23.3067	4.0978	4.3624
MSDCNN	0.5544	36.3497	3.1753	2.5996
TFNet	0.4991	37.2622	2.7884	2.3161
ResTFNet	0.4598	37.9749	2.6952	2.1537
SSFCNN	1.4151	28.2108	11.4498	7.0424
ConSSFCNN	2.5058	23.2475	15.2048	12.9631
SSR-NET	0.4106	38.9583	2.7994	2.0108
HSR-NET	0.4036	39.1085	**2.2527**	1.8955
MCT-Net	**0.3363**	**40.6931**	**2.2186**	**1.6326**
SS-INR	**0.3317**	**40.8126**	5.4441	**1.6053**

TABLE IV
EXPERIMENTAL RESULTS ON THE HOUSTON DATASET, WITH THE FIRST AND SECOND PLACE BEING, RESPECTIVELY, MARKED IN **RED** AND **BLUE**

Method	Houston			
	RMSE	PSNR	ERGAS	SAM
ICCV15	1.4718	42.1035	1.7400	2.2521
FUSE	1.3324	43.0584	2.0737	2.0569
MSDCNN	1.7692	42.7828	2.1500	3.0875
TFNet	1.3988	44.8236	1.7854	2.3991
ResTFNet	1.0959	46.9434	1.5008	1.9094
SSFCNN	3.7589	36.2373	4.5855	7.8291
ConSSFCNN	3.0424	38.0742	9.2973	6.2574
SSR-NET	1.2917	45.4764	4.0610	2.5177
HSR-NET	**0.9868**	**47.8541**	**1.3684**	**1.7393**
MCT-Net	1.3225	45.3108	1.8948	2.4096
SS-INR	**0.8071**	**49.6002**	**1.1464**	**1.3198**

TABLE V
EXPERIMENTAL RESULTS ON THE PAVIA UNIVERSITY DATASET, WITH THE FIRST AND SECOND PLACE BEING, RESPECTIVELY, MARKED IN **RED** AND **BLUE**

Method	Pavia University			
	RMSE	PSNR	ERGAS	SAM
ICCV15	2.9383	37.8965	3.2459	2.0827
FUSE	2.3200	40.2630	2.5369	**1.6827**
MSDCNN	2.3207	40.5032	1.6358	2.6309
TFNet	2.2427	40.8004	1.6053	2.4890
ResTFNet	2.0578	41.5477	1.5068	2.3517
SSFCNN	1.9410	42.0550	1.3957	2.2121
ConSSFCNN	2.3916	40.2420	1.6340	2.4235
SSR-NET	1.6447	43.4938	1.2146	1.9329
HSR-NET	1.8676	42.3924	1.3778	2.1745
MCT-Net	**1.5555**	**43.9784**	**1.1721**	1.8512
SS-INR	**1.4970**	**44.3133**	**1.1377**	**1.7989**

TABLE VI
EXPERIMENTAL RESULTS ON THE XIONGAN DATASET, WITH THE FIRST AND SECOND PLACE BEING, RESPECTIVELY, MARKED IN **RED** AND **BLUE**

Method	Xiongan			
	RMSE	PSNR	ERGAS	SAM
ICCV15	9.7870	23.3026	5.5366	3.1392
FUSE	10.8006	20.3530	6.9791	2.9049
MSDCNN	2.6058	34.3962	1.8149	1.7689
TFNet	2.2089	35.8318	1.4449	1.6029
ResTFNet	**2.0892**	**36.3156**	1.4582	1.5129
SSFCNN	4.7061	29.2620	2.6383	5.7081
ConSSFCNN	9.0627	23.5700	14.3541	9.8958
SSR-NET	3.6746	31.4109	2.0647	1.8816
HSR-NET	2.2356	35.7274	1.4594	1.6377
MCT-Net	2.1114	36.1950	**1.4130**	**1.4120**
SS-INR	**1.9973**	**36.7065**	**1.1948**	**1.3306**

datasets. The ten compared algorithms contain two traditional fusion methods (CSU [39], SSE [7]) and eight deep learning–based fusion methods (TFNet [16], ResTFNet [16], SSFCNN [40], ConSSFCNN [40], MSDCNN [41], SSR-NET [18], HSR-NET [17], MCT-NET [42]). Tables II–VI shows the metrics of these fusion methods on different datasets, and Figs. 7–11 show the visual fusion results. In order to facilitate comprehensive comparison, we presented not only the pseudo-color RGB image of the fusion result but also the difference image between the fusion result and the reference image, as well as the locally amplified image of the details. Due to space limitations, only the results of fusion methods based on deep learning are shown.

1) Results on Pavia Centre: Table II shows the objective evaluation metrics of the fusion results on the Pavia Centre dataset. Fig. 7 shows the visual results graph. Based on the experimental outcomes, it is evident that SS-INR yields the most favorable results across all four metrics. The SAM metric is reduced by 7.18% compared to MCT-Net, which is a good proof that SS-INR a good ability to reconstruct spectral information. It is also noteworthy that our SS-INR achieves satisfactory visual reconstruction results.

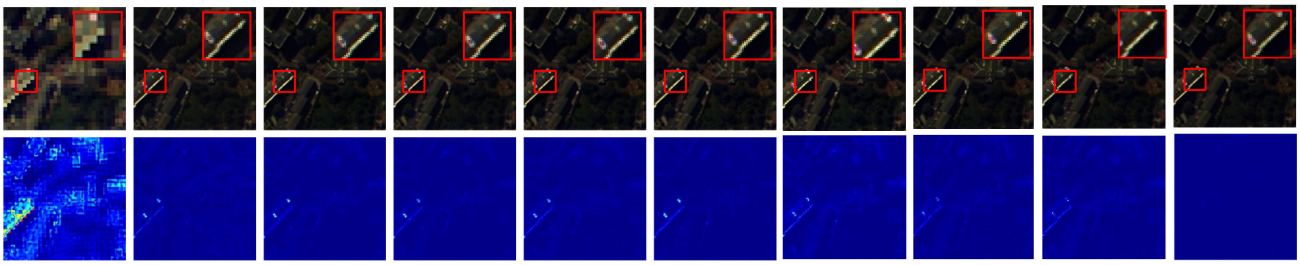
2) Results on Botswana: Table III and Fig. 8 show the objective evaluation metrics and visual comparisons of SS-INR and other fusion methods on the Botswana dataset, respectively. It can be seen that SS-INR achieves satisfactory results with the best scores on three metrics. Although the Botswana dataset has a very low spatial resolution, SS-INR still achieves

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TABLE VII

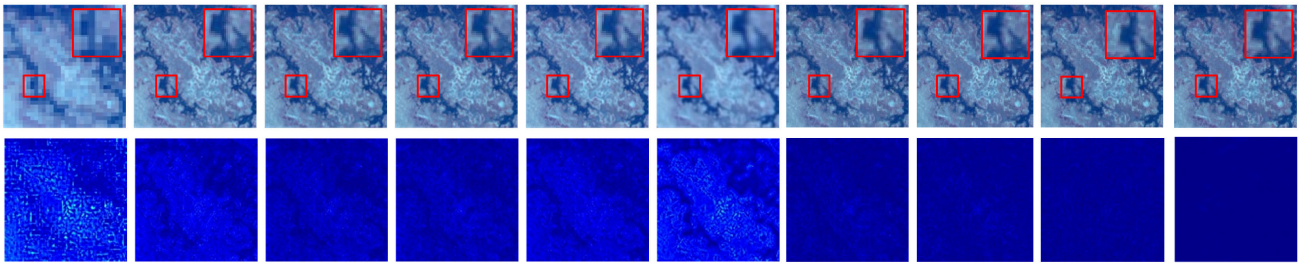
RESULTS OF ABLATION EXPERIMENTS ON SPATIAL-INR AND SPECTRAL-INR ON THE PAVIA CENTRE DATASET. THE BEST SCORES ARE MARKED IN **RED**

spatial-INR1	spatial-INR2	spectral-INR	billinear1	billinear2	Conv	RMSE	PSNR	ERGAS	SAM
	✓	✓	✓			4.1980	35.5469	4.6879	4.8633
✓		✓		✓		2.7266	49.4120	3.0874	3.6835
		✓	✓	✓		4.5913	34.7519	5.1795	5.3497
✓	✓				✓	2.6430	39.7066	3.0040	3.6544
✓	✓	✓				**2.5745**	**39.9169**	**2.9595**	**3.4244**



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 7. Fusion results of different methods on Pavia Centre dataset. The initial row describes the fused image in pseudo-color RGB format, while the second row shows the difference images between the estimated and the reference pseudo-color RGB images. (a) MSDCNN. (b) TFNet. (c) ResTFNet. (d) SSFCNN. (e) ConSSFCNN. (f) SSR-NET. (g) HSR-Net. (h) MCT-Net. (i) SS-INR. (j) Ground truth.



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 8. Fusion results of different methods on Botswana dataset. The initial row describes the fused image in pseudo-color RGB format, while the second row shows the difference images between the estimated and the reference pseudo-color RGB images. (a) MSDCNN. (b) TFNet. (c) ResTFNet. (d) SSFCNN. (e) ConSSFCNN. (f) SSR-NET. (g) HSR-Net. (h) MCT-Net. (i) SS-INR. (j) Ground truth.

TABLE VIII

RESULTS OF ABLATION EXPERIMENTS ON THE SPATIAL AND SPECTRAL BRANCHES WITHIN BPMF ON THE PAVIA CENTRE DATASET. THE BEST SCORES ARE MARKED IN **RED**

Spatial branch	Spectral branch	RMSE	PSNR	ERGAS	SAM
✗	✓	3.0213	38.6033	3.4095	3.4772
✓	✗	2.9582	38.7065	3.3583	3.4901
✗	✗	3.0944	38.2921	3.5043	3.5178
✓	✓	**2.5745**	**39.9169**	**2.9595**	**3.4244**

TABLE IX

PARAMETERS AND TEST TIME OF THE PROPOSED SS-INR AND THE OTHER DEEP LEARNING MODELS. THE BEST DATA ARE MARKED IN **BOLD**

Method	Pavia Center			
	Params (M)	Test time (ms)	PSNR	SAM
MSDCNN	1.2426	15.2216	35.7370	5.4004
TFNet	**0.2488**	13.7091	35.5686	4.8437
ResTFNet	2.3241	13.7824	36.2522	4.6073
SSFCNN	0.4523	13.6759	34.5569	5.7962
ConSSFCNN	0.4696	13.7539	33.3239	6.1196
SSR-NET	0.2812	13.7230	37.4752	3.8618
HSR-NET	3.2670	13.9562	37.4889	4.1796
MCT-Net	4.0177	**13.5350**	39.3378	3.6897
SS-INR	1.3659	42.5584	**39.8684**	**3.4250**

satisfactory results, which fully demonstrates that SSR-INR has a good ability to reconstruct spatial information in lower spatial resolution scenarios. The visual results also show that the images after SS-INR fusion have sharper edges.

3) Results on Houston: Table IV shows the evaluation metrics of the fusion results on the Houston dataset. Fig. 9 depicts the visual results after fusion. The experimental results show that SS-INR achieves the best results for all four metrics. Compared with HSR-NET, our method improves PSNR by 3.65% and reduces RMSE, ERGAS, SAM by 18.21%, 16.22%, and 24.12%, respectively. It can be seen that SS-INR has excellent spectral information reconstruction ability. The visual results show that the HSI reconstructed by SS-INR is sharper, which indicates that SS-INR is able to recover the spatial structure better.

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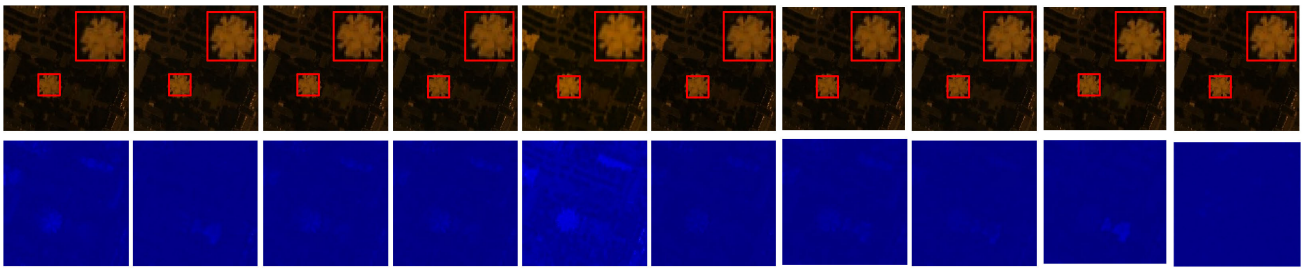


Fig. 9. Fusion results of different methods on Houston dataset. The initial row describes the fused image in pseudo-color RGB format, while the second row shows the difference images between the estimated and the reference pseudo-color RGB images. (a) MSDCNN. (b) TFNet. (c) ResTFNet. (d) SSFCNN. (e) ConSSFCNN. (f) SSR-NET. (g) HSR-Net. (h) MCT-Net. (i) SS-INR. (j) Ground truth.



Fig. 10. Fusion results of different methods on Pavia University dataset. The initial row describes the fused image in pseudo-color RGB format, while the second row shows the difference images between the estimated and the reference pseudo-color RGB images. (a) MSDCNN. (b) TFNet. (c) ResTFNet. (d) SSFCNN. (e) ConSSFCNN. (f) SSR-NET. (g) HSR-Net. (h) MCT-Net. (i) SS-INR. (j) Ground truth.

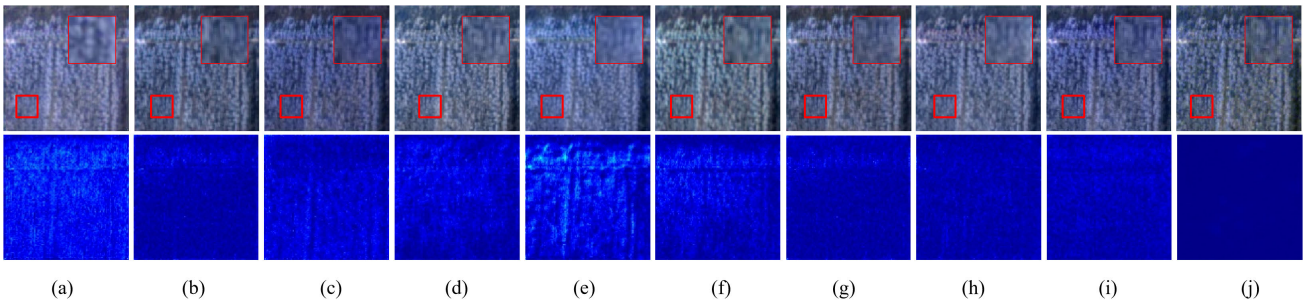


Fig. 11. Fusion results of different methods on Xiongan dataset. The initial row describes the fused image in pseudo-color RGB format, while the second row shows the difference images between the estimated and the reference pseudo-color RGB images. (a) MSDCNN. (b) TFNet. (c) ResTFNet. (d) SSFCNN. (e) ConSSFCNN. (f) SSR-NET. (g) HSR-Net. (h) MCT-Net. (i) SS-INR. (j) Ground truth.

4) Results on Pavia University: Table V shows the evaluation metrics on the Pavia University dataset, and Fig. 10 shows the visual results of all fusion methods. The experimental results show that SS-INR achieves satisfactory results in both spatial and spectral reconstructions. Despite the high spatial resolution of 1.3 m in the Pavia University dataset, SS-INR still achieves good visual results. Compared with the traditional one-way fusion structure, our method can better use the relationship shared between LR and HR images, which facilitates the recovery of detailed information.

5) Results on Xiongan: The results of the evaluation metrics of each fusion method on the Xiongan dataset are shown in Table VI, and Fig. 11 shows the graph of the fusion results of each method. The experimental results show that on the real dataset, SS-INR achieves the best performance in all four indicators. It is worth noting that although the Xiongan dataset contains up to 256 bands, SS-INR still achieves the best SAM indicator. This clearly demonstrates the excellent spectral information reconstruction ability and the robustness of SS-INR in handling high-spectral-resolution reconstruction tasks.

E. Ablation Study

To validate the effectiveness of each component contained in the proposed SS-INR, we conducted a series of ablation experiments on the Pavia Centre dataset. The ablation study can be further stratified into two individual parts. The first part focused on the spatial-INR and spectral-INR within the network, to examine the efficacy of the INR method in spatial and spectral reconstruction. The second part of the study involved ablation studies performed on the spatial and spectral branches within BPF, to confirm the feasibility of utilizing the deep back-projection network in fusion tasks. The network structure corresponding to the above experiments is described in Fig. 12, and the experimental results are shown in Fig. 13.

1) Ablation Experiments on Spatial-INR and Spectral-INR: In the traditional image representation, each pixel point represents only a discrete position in 2-D space, so its expressiveness is limited by the resolution, and the INR approach can solve this problem well. Based on this, in SS-INR, we design spatial-INR and spectral-INR to implement upsampling of images in spatial dimension and spectral dimension, respectively. To validate their effectiveness, we did three ablation

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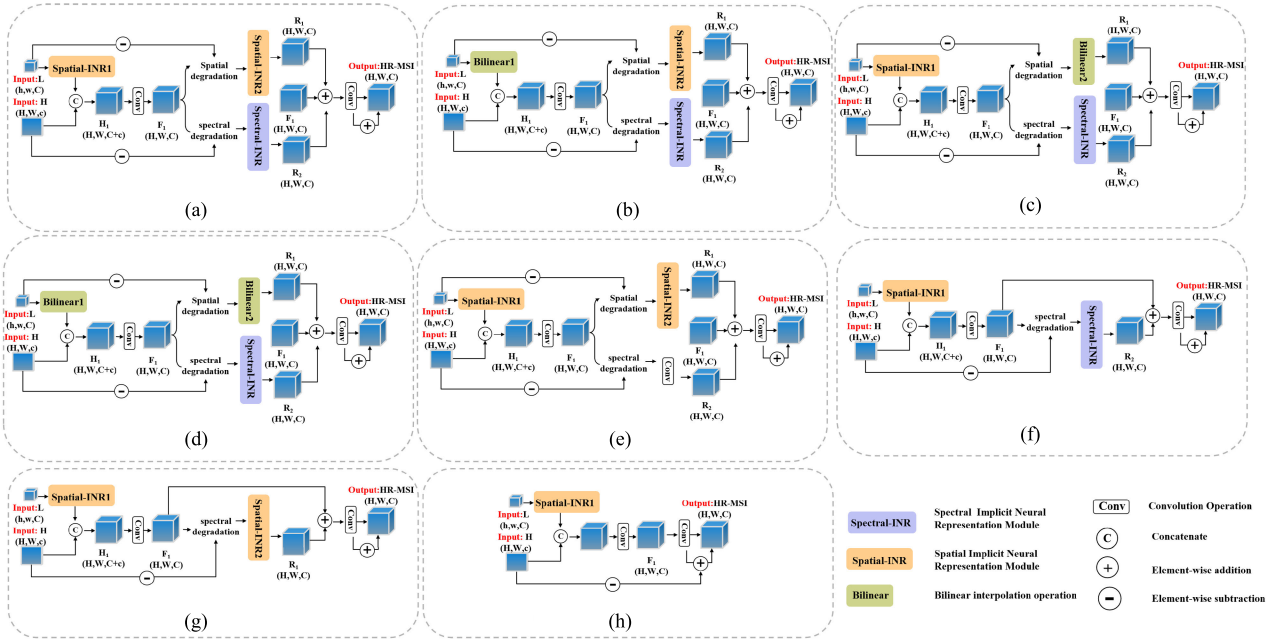


Fig. 12. Network architecture diagrams for each ablation experiment. (a) SS-INR. (b) Replacing spatial-INR1 with bilinear interpolation. (c) Replacing spatial-INR2 and spectral-INR with bilinear interpolation and convolution. (d) Replacing all spatial-INR with bilinear interpolation. (e) Replacing spectral-INR with convolution. (f) Removing the spatial branch of BPFM. (g) Removing the spectral branch of BPFM. (h) Removing the BPFM.

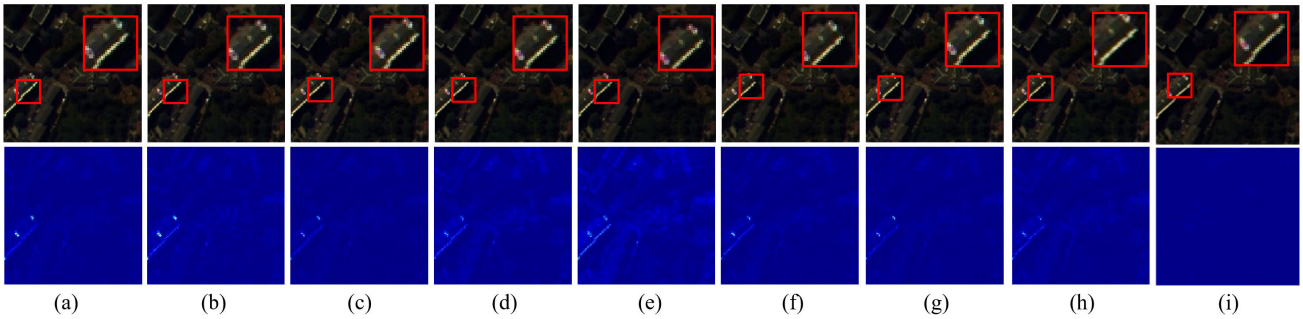


Fig. 13. Fusion results of ablation study. The first row shows the RGB result images after fusion, while the second row shows the difference images between the estimated and the reference pseudo-color RGB images. (a) Replacing spatial-INR1 with bilinear interpolation. (b) Replacing spatial-INR2 and spectral-INR with bilinear interpolation and convolution. (c) Replacing all spatial-INR with bilinear interpolation. (d) Replacing spectral-INR with convolution. (e) Removing the spatial branch of BPF. (f) Removing the spectral branch of BPF. (g) Removing the BPF. (h) SS-INR. (i) Ground truth.

experiments: 1) replacing spatial-INR1 with bilinear interpolation; 2) replacing spatial-INR2 and spectral-INR with bilinear interpolation and convolution; and 3) replacing all spatial-INR with bilinear interpolation. The network architecture diagram corresponding to the above experiments is shown in Fig. 12(b)–(e), and the experimental evaluation metrics results are shown in Table VII.

The experimental results demonstrate that the initial network structure attains optimal performance, while model efficacy declines upon substitution of either spatial-INR1 or spatial-INR2 with bilinear interpolation. In particular, the quality of the fused image significantly deteriorates when spatial-INR1 is replaced, with a reduction of PSNR by 10.95% relative to SS-INR. The worst fusion outcomes are obtained when all spatial-INRs in the network are replaced with bilinear interpolation. The above experimental results fully illustrate the effectiveness of spatial-INR. Compared with, for example, the traditional spatial upsampling method of bilinear interpolation, spatial-INR can learn spatial details and location information better and thus has a stronger representation capability, which is crucial for the image fusion task. When spectral-INR was replaced with a convolution operation, the SAM increased, which proves that spectral-INR has good spectral recovery capability. In summary, the use of spatial-INR and spectral-INR proposed based on INR can represent the spatial and spectral information of the image in a continuous manner. This continuous representation is more in line with the real situation of HSIs in the real world.

2) Ablation Experiments on the Spatial and Spectral Branches in BPF: Since most of the mainstream fusion networks are feed-forward, meaning that they are capable of learning only the mapping relationship from LR to HR images in isolation, without taking into account the ill-posed nature of the super-resolution problem. Based on this, SS-INR adopts a new fusion architecture by constructing a back projection–based BPF component. BPF consists of both spatial and spectral branches to facilitate the reconstruction of spatial and spectral information, respectively. To evaluate the impact of BPF on model performance, we conducted three ablation experiments, specifically: 1) the removal of the spatial branch; 2) the removal of the spectral branch; and 3) the removal of both the spatial and spectral branches. The results of the

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evaluation metrics are shown in Table VIII. With the removal of spatial or spectral branches, the model reconstruction effect becomes worse, and the fused images show unsmooth and blurred edges. The worst model performance was observed when both spatial and spectral branches were canceled in the network, and the PSNR decreased by 4.20%, RMSE, ERGAS, and SAM improved by 20.19%, 18.41%, and 2.73%, respectively. The results of the ablation study demonstrate the superiority of the two-branch fusion strategy used in BPF. The residual images obtained by the degraded model can not only provide the most depth prior information as feedback to guide the model for better training, but also better characterize the interrelationship between LR and HR images, which greatly improves the model performance.

F. Time Efficiency Analysis of Model

Table IX presents the parameters and test time of each method applied to the Pavia Centre dataset. Additionally, to facilitate comparison, the PSNR and SAM metrics are also provided. After learning, the INR can be mapped to arbitrary attribute values through different coordinate inputs, which has the stronger representational ability, high generalization, and ease of learning. However, INR requires computation of the function values of the entire volume's voxels, inevitably leading to the issue of large computational cost. However, from the experimental results, this increase in computation is acceptable. Although the parameters of SS-INR are smaller than those of ResTFNet and MCT-Net, better results are achieved. Of course, in the future, we will explore faster fusion structures to realize the performance-efficiency trade-off.

V. Conclusion

In this article, we propose a novel spatial-spectral INR network for HSI and MSI fusion (SS-INR). The SS-INR consists of two fusion stages: FF and BPF. Inspired by the concept of INR, spatial-INR and spectral-INR are designed for spatial and spectral resolution enhancement, respectively, to better simulate the continuous representation of spatial and spectral information in HSIs. Moreover, in the BPF, we investigate the degradation mechanisms of spatial and spectral domains and incorporate them as prior knowledge to guide the neural network optimization process. This approach can effectively reduce the possible mapping space and more fully exploit the intrinsic relationship between HR and LR image pairs. The proposed SS-INR model is evaluated on five widely used hyperspectral datasets, and the experimental results demonstrate its effectiveness. Furthermore, we expect that this work can provide some insights for other image fusion tasks, especially in remote sensing image processing.

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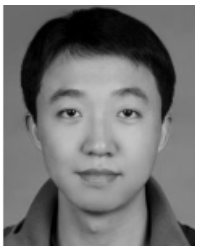
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